

CSS Phase 7: Transforms, Transitions & Animations

Introduction

CSS-এর মাধ্যমে Element-কে:

- Move করা যায়
- Rotate করা যায়
- Zoom করা যায়
- Animate করা যায়
- Smooth Effect দেওয়া যায়

এই Phase-এ আমরা শিখব:

✓ Transform

✓ Transition

✓ Animation

1. CSS Transform

Transform ব্যবহার করে Element-এর Shape বা Position পরিবর্তন করা যায়।

translate()

Element Move করার জন্য।

```
```css id="y4m8q2" .box{ transform:translate(50px, 20px); }
```

Meaning:

```
```text id="h8p3v6"
50px → Right
20px → Down
```

translateX()

```
```css id="w2n6k9" transform:translateX(100px);
```

---

### translateY()

```
```css id="p5x1r4"
transform:translateY(50px);
```

rotate()

Element Rotate করে।

```
```css id="v7c3m8" transform:rotate(45deg);
```

---

Example:

```
```css id="a1k9p5"
img:hover{
    transform:rotate(10deg);
}
```

scale()

Element বড় বা ছোট করে।

```
```css id="n4q8w2" transform:scale(1.2);
```

Meaning:

```
```text id="z6r3y7"
1.0 = Original Size
1.2 = 20% Bigger
0.8 = Smaller
```

Example

```
```css id="k8m1v4" .card:hover{ transform:scale(1.05); }
```

---

# skew()

Element কাত করে।

```
```css id="q3x7n5"
transform:skew(15deg);
```

Multiple Transform

একসাথে একাধিক Transform ব্যবহার করা যায়।

```
```css id="m7p2k8" transform: translateY(-10px) scale(1.05) rotate(5deg);
```

---

# 2. CSS Transition

Transition Element-এর পরিবর্তনকে Smooth করে।

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Without Transition:

```
```css id="r5n8x3"
button:hover{
    background:red;
}
```

Color হঠাৎ পরিবর্তন হবে।

With Transition:

```
```css id="b2m6q9" button{ transition:0.3s; }
```

Color Smoothভাবে পরিবর্তন হবে।

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# Transition Syntax

```
```css id="u9p4k7"
transition:
property duration timing-function delay;
```

Example:

```
```css id="f1x8n2" transition: all 0.3s ease;
```

---

# Common Timing Functions

### ease

```
```css id="j6m3v8"
transition:0.3s ease;
```

linear

```
```css id="w8p2q5" transition:0.3s linear;
```

Constant Speed।

---

### ease-in

```
```css id="s4n7k1"
transition:0.3s ease-in;
```

Slow Start।

ease-out

```
```css id="x5m9p3" transition:0.3s ease-out;
```

```
Slow End|
```

```

```

```
ease-in-out
```

```
```css id="c2q8v6"  
transition:0.3s ease-in-out;
```

Most Used |

Hover Effect Example

```
```css id="t7p4m2" .button{
```

```
background:blue;

color:white;

padding:12px 20px;

transition:0.3s;
```

```
}
```

```
.button:hover{
```

```
background:red;

transform:scale(1.1);
```

```
} ```
```

---

## Card Hover Effect

```
css id="v1k7q8"
.card{
 transition:0.3s;
}
.card:hover{
 transform:translateY(-10px);
 box-shadow:
 0 10px 20px rgba(0,0,0,.2);
}
```

---

## 3. CSS Animation

Transition-এর তুলনায় Animation অনেক Powerful।

Animation নিজে নিজেই চলতে পারে।

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## @keyframes

Animation-এর Rules তৈরি করে।

```
css id="p8m2x6" @keyframes move{
```

```
 from{
 transform:translateX(0);
 }
 to{
 transform:translateX(200px);
 }
```

```
} ``
```

---

# Applying Animation

```
css id="k4n8v1"
.box{
 animation:move 2s;
}
```

Meaning:

```text id="r7m5q3" move = Animation Name 2s = Duration

```
---

# Infinite Animation

```css id="n2x6p8"
animation:
move 2s infinite;
```

বারবার চলবে।

---

## Animation Direction

```css id="w4m9k2" animation-direction: alternate;

```
Forward + Backward চলবে।

---

# Animation Delay

```css id="j8p3x7"
animation-delay:2s;
```

2 সেকেন্ড পরে শুরু হবে।

---

# Animation Timing Function

```
```css id="u5k1m9" animation-timing-function:ease;
```

```
---
```

```
# Complete Animation Example
```

```
HTML:
```

```
```html id="d9q7v4"
<div class="box"></div>
```

CSS:

```
```css id="a3m8k5" .box{
```

```
width:100px;
```

```
height:100px;
```

```
background:red;
```

```
animation:
```

```
bounce 2s infinite alternate;
```

```
}
```

```
@keyframes bounce{
```

```
from{
```

```
    transform:translateY(0);
```

```
}
```

```
to{
```

```
    transform:translateY(-100px);
```

```
}
```

```
}``` Result: Box উপরে নিচে Bounce করবে।
```

Loading Spinner Example

```
css id="k6x2p9"
.loader{

    width:50px;

    height:50px;

    border:5px solid gray;

    border-top-color:black;

    border-radius:50%;

    animation:
    spin 1s linear infinite;
}

@keyframes spin{

    from{
        transform:rotate(0deg);
    }

    to{
        transform:rotate(360deg);
    }
}
```

Most Important Properties

Transform

```css id="f4m7p2" translate()

translateX()

translateY()

rotate()

scale()

skew()

```

Transition

```css id="x1k9m8"  
transition  
  
duration  
  
timing-function  
  
delay
```

Animation

```
```css id="v8p2q4" @keyframes
```

animation-name

animation-duration

animation-delay

animation-direction

animation-iteration-count

animation-timing-function ```

---

## Real Project Ideas

### Beginner

1. Animated Button
2. Hover Card
3. Zoom Image

### Intermediate

1. Loading Spinner

- 2. Animated Navbar
- 3. Pricing Card Hover Effect

### Advanced

- 1. Animated Landing Page
  - 2. Hero Section Animation
  - 3. Interactive Dashboard UI
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## Phase 7 Checklist

- ☐ translate()
- ☐ translateX()
- ☐ translateY()
- ☐ rotate()
- ☐ scale()
- ☐ skew()
- ☐ Multiple Transform
- ☐ transition
- ☐ duration
- ☐ timing-function
- ☐ delay
- ☐ Hover Animation
- ☐ @keyframes
- ☐ animation-name
- ☐ animation-duration
- ☐ animation-delay
- ☐ animation-direction

☐ infinite

☐ Loading Spinner

☐ Animated Button

☐ Hover Card

Phase 7 শেষ হলে তোমার Website Static থাকবে না। User Interaction-এর সময় Smooth Effect, Hover Animation এবং Professional UI Experience তৈরি করতে পারবে।